

BUS 696 Final Project: Pitch Your Trading Strategy

Professor Jonathan Hersh | Spring 2026 | Chapman University Weight: 35% of final grade Due: Week 13 (May 13) — Presentations in class + Notebook due by 11:59pm Groups of 2-3

The Scenario

You are a quantitative trading team pitching to a skeptical investor. You've built an AI-driven trading strategy and you want their capital. Your job is to:

1. **Build the strategy** in a Jupyter notebook — data, model, backtest, risk management
2. **Present it as a pitch** — convince the investor your edge is real, your risk management is sound, and you deserve their money

The investor (your professor and classmates) has read Lopez de Prado cover to cover. They know every trick in the book — overfitting, look-ahead bias, cherry-picked time periods, ignoring transaction costs, data snooping. They will ask hard questions. You need to have answers.

What You're Building

A complete trading strategy that incorporates the key concepts from this course. Your notebook and presentation must address all of the following:

Data & Features

- Financial data from any market (stocks, ETFs, crypto, prediction markets — your choice)
- At least **5 engineered features**, all properly lagged (no look-ahead bias)
- **All signals must be lagged by at least one period** unless you provide an explicit justification for why real-time access is realistic
- Features should represent **distinct economic hypotheses**, not minor variations of the same transformation (e.g., five moving averages of different lengths counts as one idea, not five)
- **Clearly define your prediction target** (e.g., next-day return, next-week direction, next-month rank). Your target horizon must match your trading frequency — don't predict monthly returns and trade daily
- You must **clearly articulate the economic intuition behind your signal**. Why should this pattern exist? Purely statistical patterns without a plausible mechanism will be viewed skeptically. "XGBoost found it" is not an explanation.
- Clear documentation of data sources

ML Model

- At least **one ML model** from the course (logistic regression, Lasso, random forest, XGBoost, DNN, LSTM, or LLM-based)
- Comparison to at least **two baselines**: (1) a naive strategy (e.g., buy-and-hold, always predict majority class, random), and (2) a simple model (e.g., logistic regression). This lets you measure whether your model adds value beyond doing nothing *and* beyond a simple approach.

Walk-Forward Backtesting

- **Walk-forward validation** with expanding or rolling training windows — not a single train/test split
- **Transaction costs** included (state and justify your assumptions)
- All reported results must be **out-of-sample**

Minimum backtest spec (to prevent ambiguity):

Requirement	Minimum
Walk-forward folds	At least 5 (e.g., test years 2020-2024)
Rebalancing frequency	Explicitly stated and justified
Signal -> position mapping	Clearly defined (how does a prediction become a trade?)
Transaction cost assumption	Stated in basis points with source/justification
Turnover reported	Report annualized turnover. High-turnover strategies must justify that costs are realistic.
No overlapping labels	Training labels must not overlap with test period

Risk Management

This is where the investor pays the most attention.

- **Position sizing**: Use the Kelly criterion. Show how you estimate the edge (expected return or win probability) that feeds into the Kelly formula. If you use ML-predicted probabilities, discuss whether they are **well-calibrated** — if your model says 60% but the true rate is 52%, your Kelly sizing is dangerously wrong.
- **Drawdown analysis**: What is the maximum drawdown? How long does it take to recover? Would the investor be able to stomach it?
- **Regime awareness**: How does your strategy perform in calm markets vs. volatile/crisis markets? Does it blow up exactly when the investor needs it most?

Robustness: Kill Your Strategy

Show at least **one test where your strategy's performance degrades**. This is mandatory.

Examples: - Add an extra day of lag to your signals - Double your transaction cost assumption - Test on a different time period or market - Remove your best-performing feature

If you can't find anything that hurts performance, you haven't looked hard enough (or something is wrong).

Honest Assessment

- **vs. Buy-and-hold:** Show a clear comparison. If you can't beat buy-and-hold after costs, say so and explain why.
- **Multiple testing:** How many choices did you make (model, features, hyperparameters, universe, rebalancing frequency)? Each is an implicit test. Address the data snooping risk.
- **Lopez de Prado critique:** Which of his "10 Reasons Most ML Funds Fail" apply to your strategy?
- **EMH connection:** What form of market efficiency would need to be violated for your strategy to work?

Markets

Trade whatever you want. Stocks, bonds, crypto, prediction markets, commodities, FX, volatility products — if you can get the data and make a case for an edge, go for it. Bet on your grandma's bingo game if you can make money on it. I don't care what the market is. I care that the methodology is sound.

Some starting points:

Market	Data Source	Notes
U.S. equities	yfinance	Single stocks, sector baskets, or cross-sectional ranking
ETFs / indices	yfinance	SPY, bonds, sectors, international
Crypto	yfinance or exchange APIs	Higher volatility, 24/7 trading, different cost structure
Prediction markets	Polymarket Gamma API	Binary contracts — use KL divergence and Kelly from Week 8
Multi-asset	Mix of above	Rotation or allocation across asset classes
Anything else	You tell me	Commodities, FX, volatility (VIX), REITs — just get the data and justify the costs

Alternative Data (Bonus Points)

Incorporating alternative data earns **up to 10 bonus points** (out of 100). This could include:

- LLM-extracted news sentiment (Claude API)
- Google Trends search data
- Social media signals
- Macro indicators from FRED
- Insider trading filings (SEC Form 4)

- Earnings surprise data

To earn the bonus: 1. Show whether the alt data **adds predictive power** beyond price/volume features
 2. Compute the signal's **Information Coefficient (IC)** 3. Test for **signal decay** — does the signal still work with a longer lag?

Deliverables

1. The Pitch Presentation (40% of project grade)

15-20 minutes in class on May 13, followed by **5-10 minutes of adversarial Q&A**.

You are pitching to an investor. Structure your presentation as:

The Setup - What is your strategy? What market, what signal, what edge? - Why should this edge exist? (Behavioral bias? Structural friction? Information asymmetry?)

The Evidence - Walk-forward backtest results (out-of-sample only) - Key performance metrics: annualized return, Sharpe ratio, max drawdown - How it compares to buy-and-hold

The Risk Management - How you size positions (Kelly) and why - Drawdown analysis — worst-case scenario - Performance across market regimes - What happens if your model is wrong?

The Honest Take - What are the weaknesses? What could go wrong? - Your “kill your strategy” test — what breaks it? - Multiple testing / data snooping concerns - What would you do with 6 more months?

The Ask - How much capital do you want? - What return can the investor realistically expect? - What is the investor's worst-case scenario?

Be prepared to defend every modeling choice in Q&A. The professor and classmates will challenge you on overfitting, costs, assumptions, and whether your edge is real. Groups that can handle tough questions score significantly higher.

Every group will be asked one final question: **“Would you invest your own \$50,000 in this strategy? Yes or no.”** There's no right answer — but your reasoning matters.

Presentation grading:

Criterion	Weight
Clarity — can a non-technical investor follow?	25%
Depth — can you answer tough questions about methodology?	25%
Honesty — do you confront the hard truths?	25%
Visuals — clear, labeled, not cluttered?	25%

2. Jupyter Notebook (50% of project grade)

A single .ipynb file that contains all your code and analysis. It must **run end-to-end and reproduce your results**. If your notebook doesn't run, you lose 10% automatically — this mirrors real research standards where replication is non-negotiable.

Include markdown sections explaining your choices. We're looking for:

Component	Weight	Full Credit
Data & features	10%	Distinct hypotheses. No look-ahead bias. Clear docs.
Model	10%	Appropriate model. Baseline comparison. Hyperparameters justified.
Backtesting	25%	Walk-forward (5+ folds). Transaction costs. OOS results only.
Risk management	25%	Kelly sizing with calibrated edge estimate. Drawdown analysis. Regime awareness.
Robustness test	10%	At least one test showing performance degradation.
Honest assessment	20%	Buy-and-hold comparison. Multiple testing. LdP critique. EMH.
Alt data bonus	+10	IC analysis. Decay test. Adds value beyond price/volume.

3. Peer Evaluation (10% of project grade)

Confidential form rating each team member's contribution.

Grading Philosophy

Honesty is rewarded. Bullshit is not.

A strategy that doesn't beat the market can earn full marks if: - The backtesting is rigorous (walk-forward, costs included) - The risk management is thoughtful (Kelly, drawdown analysis) - The robustness test is genuine (you actually tried to break it) - The assessment is honest ("here's why it doesn't work, here's what we learned")

A strategy that claims to beat the market will be scrutinized: - Is it walk-forward or a single split? - Are costs included? - How many design choices were made? (data snooping risk) - Does the Sharpe ratio survive regime analysis? - Would Lopez de Prado approve?

Too-good-to-be-true penalty: Strategies reporting Sharpe ratios above 2.0 or annualized returns above 30% without a compelling, well-documented justification will receive additional scrutiny and may be penalized. Extraordinary claims require extraordinary evidence.

Timeline

Date	Milestone
Week 9 (April 15)	Form groups. Submit members + 2-sentence strategy idea via Canvas.
Week 10 (April 22)	Submit 1-page proposal: market, data, planned model, evaluation approach.
Week 13 (May 13)	Presentations in class. Notebook due 11:59pm.

Tips

- **Start simple.** Get data -> model -> backtest working first. Add complexity later.
- **Reuse lab code.** You have working walk-forward backtests, Kelly functions, IC calculations, and dashboards from Weeks 5-8.
- **Cache your data.** Save downloads to CSV so your notebook doesn't break if an API is down.
- **Be skeptical of good results.** A 2.0 Sharpe should trigger "what did I do wrong?" not "I'm going to be rich."
- **Practice the pitch.** The presentation is 40% of the grade. Rehearse. Do a dry run with friends.

FAQ

Q: Can we use AI tools to help write code? A: Yes. You must understand and explain every line. We will ask during Q&A.

Q: What if our strategy doesn't work? A: That's fine. Explain why, and what it teaches us about markets.

Q: How much data do we need? A: At least 5 years for walk-forward backtesting.

Q: Can two groups use the same market? A: Yes, but your thesis and approach should differ.

Q: Can we use paid data? A: Stick to free sources unless you already have access. Ask if unsure.

"Everyone has a plan until they see their maximum drawdown."